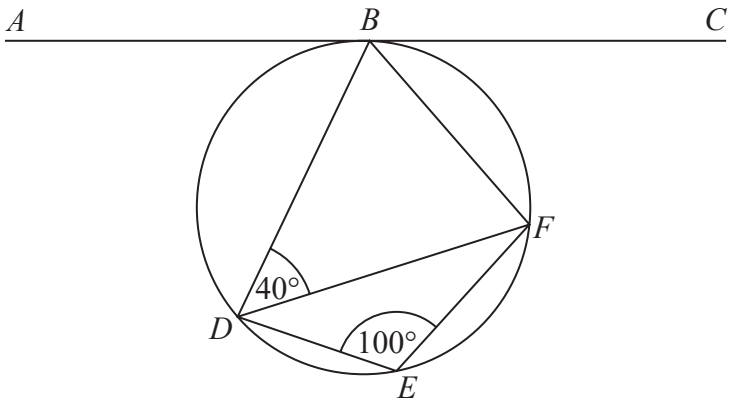
14



Points B, D, E and F lie on a circle.
ABC is the tangent to the circle at B.

Find the size of angle ABD.
You must give a reason for each stage of your working.

Angle DBF = 180° – 100° = 80° (opposite cyclic angles).

Angle BFD = 180° – 80° – 40° = 60° (triangle sum).

Angle ABD = angle BFD = 60° (alternate segment theorem).

4 marks total
Cyclic opposite angles
Triangle angle sum
Alternate segment
60° with reasons

(Total for Question 14 is 4 marks)

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